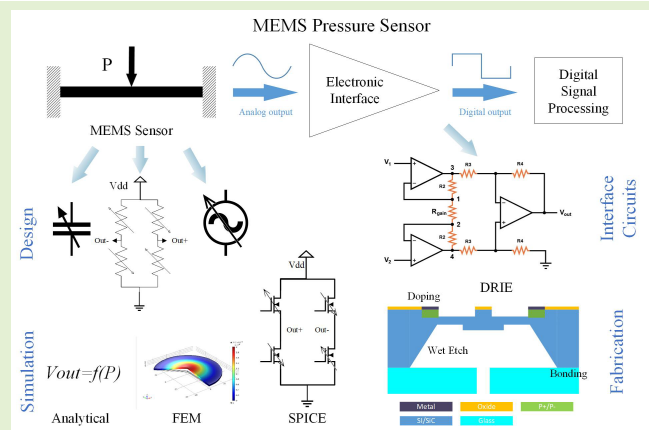


MEMS Pressure Sensors Design, Simulation, Manufacturing, Interface Circuits: A Review

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Abstract—This article deals with the fabrication technologies and design techniques of pressure sensors using various sensing element (SE) materials. The comparative analysis of different reading techniques of output signal and processing circuits was discussed. Promising approaches to design and develop microelectromechanical systems (MEMS) pressure sensors were presented. It has been proven that the main design phase is input characteristics simulation for pressure sensors that are carried out by means of numeric methods. The design techniques application (simulation) enables researchers to define the output characteristics of MEMS pressure sensors such as sensitivity, linearity, and the measured pressures range.

Index Terms—Application specific integrated circuit (ASIC), CMOS, membrane, microelectromechanical systems (MEMS), MOSFET, piezoresistive, pressure sensor, SiC, simulation.



I. INTRODUCTION

A PRESSURE sensor is defined as a device with changing parameters, which change depending on the pressure of the measured environment, and converts pressure into a proportional output electric signal.

Pressure sensors based on microelectromechanical systems (MEMS) technologies are currently applied in instrumentation and control system units in such industries as shipbuilding, automobile manufacturing, aircraft engineering, gas, petrochemical, and food industries. The evolution of micromechan-

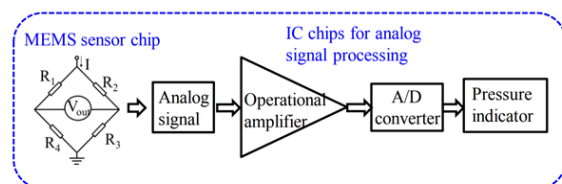


Fig. 1. Standard MEMS pressure sensor design [2].

ical pressure sensors (MMPSSs) dates back to 1954 when the piezoresistive effect in silicon and germanium was discovered [1]. A standard MEMS pressure sensor (see Fig. 1) consists of a sensing element (SE), a membrane, and an electronic processing circuit, i.e., an application-specific-integrated circuit (ASIC) [2].

The main technical characteristics of MMPSSs are as follows: the operational measurements range, the sensitivity to the measured pressure, and the output voltage. MMPSSs can be divided into absolute (measuring the absolute pressure), differential (measuring pressures differences), relative (measuring excessive pressure to the atmospheric pressure), and vacuum (measuring the degree of the subatmospherical pressure) based on the measured pressure type. Modern MEMS sensors differ significantly from the first-generation MEMS sensors as the fabrication technologies, processing techniques, and design methods have improved significantly. This research aims at reviewing modern trends in design and technologies of MEMS

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piezoresistive pressure sensors and their further improvement. The outline of the article is as follows: Section II (that follows the Introduction) describes the main approaches to design SEs of MEMS pressure sensors. Section III presents the information about the piezoresistive [3] signal detection from the SE. Section IV deals with the main technological peculiarities of the manufacturing technologies of MEMS pressure sensors, that includes fabrication and packaging. Section V deals with the promising approaches to design and develop pressure sensors in terms of novel materials and signal interface circuits (ICs). Section VI provides design directions for MEMS pressure sensors as a system based on a crystal (system-on-chip), including interface-integrated circuits and information about commercially available sensors. The conclusion presents the overall results and recommendations.

II. SIMULATION OF MEMS PRESSURE SENSORS

When designing MEMS pressure sensors, there can be several main approaches: analytical design techniques [4], [5], [6], [7], the finite element methods (FEMs) simulations using such software packages as *COMSOL Multiphysics* and *ANSYS* [8], [9], [10], [11], [12], [13], [14], [15], and other specialized software platforms to design micromechanical devices [5], [7], [16], [17], [18], for example, *Coventor* or *Intellisuite*.

The analytical design techniques of pressure sensors based on both electrostatic [5] and piezoresistive [4], [6], [7] principles can be based on making a mathematic model of a SE based on the differential equation of the thin membrane deflection as follows [6]:

$$\frac{\partial^4 w}{\partial x^4} + \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4} = \frac{P}{D} \quad (1)$$

where P —applied pressure, D —bending rigidity of thin plate (2), and w denotes small deflections of the plane in z direction [6]

$$D = \frac{Eh^3}{12(1 - \nu^2)} \quad (2)$$

where E is the modulus of elasticity of the material, ν is the Poisson's ratio, and h is the thickness of a plate.

Depending on the selecting reading method, the applied equation solutions (1) can be the cases of the maximal membrane deflections from the stable position, as well as in case of maximal mechanical stresses in the membrane. Several researchers also studied the cases of maximal average deflection and the average mechanical stress [6].

The peculiarities of the methodology consist in a possible obtaining of rapid assessment results. In general, the obtained solutions will not take into account the mechanical stress distribution along the membrane forming a parallel-plane capacitor, etc. In case, there are additional design electromechanical elements on a membrane or application of different materials in a membrane structure, the analytical descriptions can become rather complicated [19].

The FEM of design presupposes designing an FEM model of a SE of a pressure sensor with the use of special software platforms (*COMSOL* and *ANSYS*), enabling to apply the numeric solution of differential equations. The initial data

to design FEM can become the data gained at the stage of obtaining analytical evaluations.

The peculiarities of the FEM method enable design and simulation of micromechanical sensors with different types of constructive elements [19], [20], [21] and multiphysical interactions [9], [10] without the necessity of searching for an accurate analytical solution. The drawbacks of the method can be the necessity to design a reliable finite element mathematical model, as well as the complexity of integrating the developed model at the system level to assess the system as a whole.

To solve these problems, a technique using specialized software for micromechanics (*Coventor* and *Intellisuite*) can be applied [5], [7], [16], [17], [18], thus facilitating the designer to use libraries of standard structural micromechanical elements linked to the technological library, combining them into a whole shared micromechanical system.

The existing mathematical simulations of the sensor have a high degree of compliance with the finite element model and allow carrying out simulations, including the simulations that are made at the system level.

III. PIEZORESISTIVE MEMS SEs OF PRESSURE SENSORS

The main part of a SE of a MMPS is its membrane [22]. External pressure causes mechanical stress inside the membrane, which causes it to deviate. To gain information about mechanical impact on a membrane in MMPS, different methods of registering important data can be used. Piezoresistive, piezoelectric, and frequency methods can be used to measure different types of mechanical stresses [23]. The capacitive, optical, and inductive [24] methods are employed to measure the membrane displacement.

The majority of researchers study pressure sensors based on the piezoresistive registering method using one or two bridge connections of integrating piezoresistors, located on a SE [4], [6], [7], [12], [13], [14], [17], [18], [19], [20], [21], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48].

The SEs of the piezoresistive sensor (piezoresistors) are fabricated on the membrane diaphragm. The standard MMPS [4], [6], [18], [25], [33], [39], [43], [46] based on piezoresistors is shown in Fig. 2. The geometric sizes of the membrane are often chosen taking into account the small deformation theory of a thin plate as follows [25]:

$$w_{\max} = 0.0152 \frac{PL^4(1 - \nu^2)}{Eh^3} \leq \frac{1}{5}h \quad (3)$$

where $w_{\max}$ is the maximum deflection of the membrane and L is the membrane side length.

The change of piezoresistor resistance that is proportional to deformation provides the measurement of the applied pressure. The resistors are connected into the Wheatstone Bridge circuit (see Fig. 3), which allows an accurate measurement of resistance changes.

Piezoresistive elements are typically placed to incur transverse direction deformation (one part elongates, the other part contracts) to achieve the maximal output signal with the

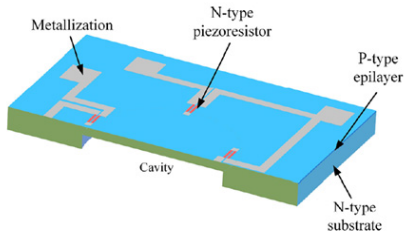


Fig. 2. Standard design of the SE of piezoresistive pressure sensor [43].

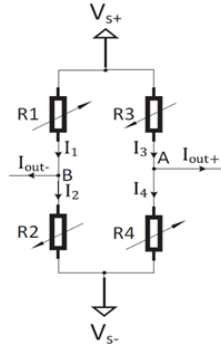


Fig. 3. Wheatstone Bridge circuit [18].

required pressure. The voltage at the bridge output can be calculated as follows [18]:

$$V_{\text{out}} = V_A - V_B = \left(\frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2} \right) V_s \quad (4)$$

where V_{out} is the voltage difference across the bridge output voltage nodes V_A and V_B , R_1 – R_4 are the corresponding bridge resistances, and V_s is the supply voltage.

Taking into account (4), it can be noted that an increase in sensor sensitivity can be achieved by maximizing the change in resistance ΔR of each piezoresistor that can be calculated as follows [35]:

$$\frac{\Delta R}{R} = \frac{\Delta l}{l} - \frac{\Delta A}{A} + \frac{\Delta \rho}{\rho} \quad (5)$$

where $\Delta R/R$ is the change in resistance, $\Delta l/l$ is the change in resistor length, $\Delta A/A$ is the change in resistor cross section, and $\Delta \rho/\rho$ is the change in resistivity due to induced mechanical stress. The resistance change $\Delta R/R$, in its turn, depends significantly on the change in specific resistivities in longitudinal $(\Delta \rho/\rho)_l$ and transverse $(\Delta \rho/\rho)_t$ directions calculated from the following equations [6], [35]:

$$(\Delta R/R)_l = (\Delta \rho/\rho)_l = \pi_l \sigma_l + \pi_t \sigma_t \quad (6)$$

$$(\Delta R/R)_t = (\Delta \rho/\rho)_t = \pi_l \sigma_l + \pi_t \sigma_t \quad (7)$$

where $(\Delta R/R)_l$ and $(\Delta R/R)_t$ are the resistance change in longitudinal and transverse direction, correspondingly. The π_l and π_t are the longitudinal and transverse piezoresistive coefficients, respectively. The σ_l and σ_t are the longitudinal and transverse mechanical stresses acting on resistors, respectively.

As it is known [1], [3], silicon piezoresistive characteristics are anisotropic; thus, the direction of piezoresistors [110] on a silicon plate (100) is used to maximize longitudinal and transverse piezoresistive coefficients. In this case, the

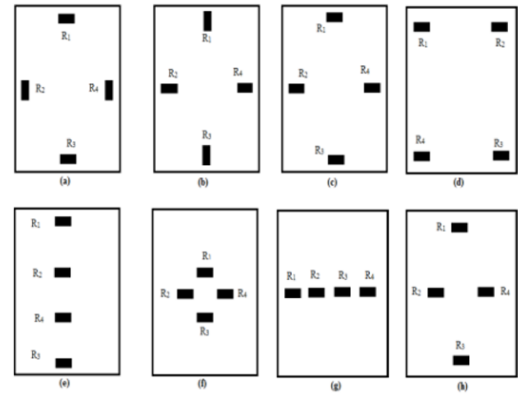


Fig. 4. Standard topologies of piezoresistors for different membrane designs [38]. (a)–(h) Placement pattern types.

piezoresistors direction has a displacement of 45° of relative crystallographic direction of a silicon plate, and longitudinal π_l and transverse π_t piezoresistive coefficients can be calculated as follows [31]:

$$\pi_l = \frac{\pi_{11} + \pi_{12} + \pi_{44}}{2} \approx \frac{\pi_{44}}{2} \quad (8)$$

$$\pi_t = \frac{\pi_{11} + \pi_{12} - \pi_{44}}{2} \approx -\frac{\pi_{44}}{2} \quad (9)$$

where π_{11} , π_{12} , and π_{44} is the material (Si) piezoresistive coefficients [1], [3].

The increase in longitudinal and transverse mechanical stresses can be achieved by means of optimal arrangement of the piezoresistors on the membrane diaphragm. The optimal arrangement of the resistors for the square shaped membrane can be obtained from the following equations that were calculated from the differential equation (1) [6]:

$$\sigma_x = \frac{Pb^4}{(a^4 + b^4)} \left(\frac{a}{h} \right)^2 \left(\left(1 - 3 \left(\frac{x}{a} \right)^2 \right) \left(1 - 3 \left(\frac{y}{b} \right)^2 \right)^2 + v \left(\frac{a}{b} \right)^2 \left(1 - 3 \left(\frac{x}{a} \right)^2 \right)^2 \left(1 - 3 \left(\frac{y}{b} \right)^2 \right) \right) \quad (10)$$

$$\sigma_y = \frac{Pb^4}{(a^4 + b^4)} \left(\frac{a}{h} \right)^2 \left(v \left(1 - 3 \left(\frac{x}{a} \right)^2 \right) \left(1 - 3 \left(\frac{y}{b} \right)^2 \right)^2 + \left(\frac{a}{b} \right)^2 \left(1 - 3 \left(\frac{x}{a} \right)^2 \right)^2 \left(1 - 3 \left(\frac{y}{b} \right)^2 \right) \right) \quad (11)$$

where $2 \cdot b$ and $2 \cdot a$ are the plate length and width and σ_x and σ_y are mechanical stresses induced in x and y directions. Taking into account (10) and (11), the optimal arrangement of resistors x and y equals $\pm a$ and $\pm b$. Fig. 4 shows possible standard topologies of piezoresistors depending on the membrane design [36]. The most frequently used configurations are a , b , c (standard) [4], [6], [18], [19], [25], [33], [39], [46] and g , e (Double boss—membrane) [17], [26], [35], [38].

In case of piezoresistive registering, the majority of researchers propose to form piezoresistors by means of p - or n -doping of certain membrane areas. Such a piezoresistors

design is the simplest for integrating into the standard technological process. The range of the resistors can be selected from the range of several units to tens units of kilohm [19], [39] based on the required linearity correlations, consumed power, and sensitivity.

The option to simulate the doping process in *COMSOL* to determine the piezoresistors parameters based on the impurities concentration in the material is considered in [8]. The standard concentration of impurities in piezoresistors of MMPS makes up 10^{17} – 10^{19} m^{-3} . This can be explained by the fact that the resistivity decreases as the doping concentration increases, resulting in a reduction piezoresistors' value and a decrease in the MMPS sensitivity. On the other hand, as the resistance increases the power of flicker and Johnson noise increases. These factors lead to an extremum in the dependence of the signal-to-noise ratio upon the doping concentration [4]. Also, one of the most popular materials to fabricate piezoresistors is graphene and its structures [14], [15], [44], [46], [47] due to higher sensitivity and smaller impact of temperature on its characteristics.

One of the most important factors limiting the accuracy and performance of a pressure sensor is the temperature error arising due to changes in the piezoresistive coefficient. Gupta et al. [37] and Basov [48] differ several types of temperature errors: temperature output voltage offset and temperature coefficient of span caused by changes in the properties of piezoresistors, the presence of residual stresses in the design of the pressure sensor, and temperature errors caused by the presence of hysteresis. Errors caused by the presence of temperature hysteresis are also highlighted separately.

In the general case [4], [14], [34], [49], the temperature coefficient of span decreases with increasing temperature due to the fact that the piezoresistive coefficient depends on the sensor temperature and the doping level of silicon. Also, due to manufacturing technological variations of piezoresistors, the thermal resistance coefficient (TCR) and temperature gradient across the sensor a temperature-dependent voltage offset occurs. Nag et al. [47] considered an analytical expression for the sensor output signal taking into account the temperature influence on the piezoresistive coefficient and the output voltage offset [49] (12)

$$\frac{V_{\text{out}}(T)}{V_{\text{in}}} = V_{\text{offset}}(T) + \frac{1}{2} \pi_{44}(N_0, 300 \text{ K}) P(N, T) (\sigma_l - \sigma_t) \quad (12)$$

$$P(N, T) = \frac{300}{T} \frac{1}{\left(1 + \exp\left(-\frac{E_f}{K_b T}\right)\right) \ln\left(1 + \exp\left(\frac{E_f}{K_b T}\right)\right)} \quad (13)$$

where $V_{\text{out}}(T)$ is the output temperature dependent voltage, $V_{\text{in}}(T)$ is the input voltage, $V_{\text{offset}}(T)$ is the output voltage induced by TCR, the $\pi_{44}(N_0, 300 \text{ K})$ is the piezoresistance coefficient at room temperature, and $P(N, T)$ is the piezoresistance factor, which depends on the doping level N and temperature T and can be expressed by (13) where E_f is the Fermi energy of the doped silicon and K_b is the Boltzmann constant.

Compensation techniques with these types of errors include additional common sensor calibration using temperature

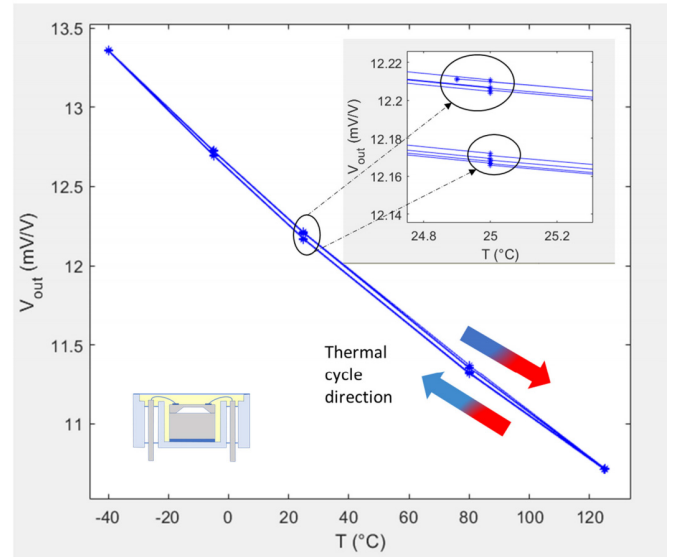


Fig. 5. Voltage output response of the sensor subjected to multiple thermal cycles [52].

dependent polynomials [49], thermodynamic control system [50] or novel algorithms based on artificial neural networks [51] or genetic algorithm [4].

A special mention is made for the temperature hysteresis error when the sensor signal differs at the same temperature and pressure depending on the temperature point from which the temperature cycle started. Hamid et al. [52] of the study showed that the greatest contribution to temperature hysteresis is made at the stage of sensor manufacturing, with further packaging contributes a relatively small component. The researchers attribute this effect to the presence of metallization, wire bonds and different coefficient of thermal expansion in the metal–silicon–glass structure. Fig. 5 shows the voltage output response of the packaged sensor with thermal hysteresis.

The main ways to deal with the thermal hysteresis phenomenon are the use of thermal cycling [52], reducing the amount of metallization on the SE [53], and building hysteresis models for possible compensation techniques [54].

IV. MANUFACTURING TECHNOLOGY OF MEMS PRESSURE SENSOR

Numerous researchers have considered standard technologies based on silicon as a membrane fabrication technology [4], [5], [6], [7], [8], [18], [19], [20], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [55], [56] as well as the silicon technology based on an insulator [silicon on insulator (SOI)] [17], [21], [38], [39], [40] or integration into a standard CMOS process [2], [8], [9], [10], [16], [48], [57], [58], [59], [60], [61], [62]. P-doped silicon plates that are applied when fabricating ASIC are used as the basis for fabricating MEMS structure. The fabrication process of micromechanical sensors consists of standard technological microelectronic procedures, but differs depending on the sensor design [63]. The typical manufacturing route based on CMOS technology consists of the following procedures [4], [19], [20], [31], [35] and is presented in Fig. 6.

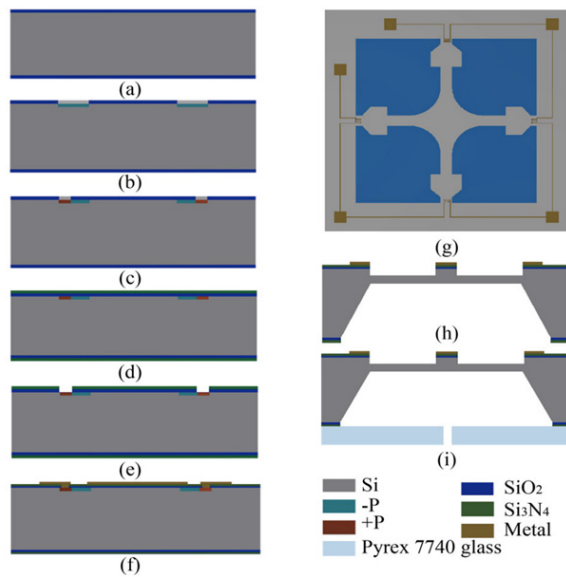


Fig. 6. Typical processing route based on CMOS technology [19]. (a) SiO_2 growth by thermal oxidation. (b) Boron ion implantation to form piezoresistors. (c) Additional boron ion implantation for heavily doped ohmic contact. (d) SiO_2 and Si_3N_4 films deposition for insulation. (e) Films lithography and etching for contact windows. (f) Metal sputtering and patterning for electrical connection of Wheatstone bridge. (g) Top silicon dry etching to form structures. (h) Anisotropic wet silicon etching from sensor bottom. (i) Anodic bonding to borosilicate glass.

- 1) Dry or liquid oxidation of a plate.
- 2) Photolithography, ion-implant doping for applying resistive strain sensors and contact windows.
- 3) Si_3N_4 deposition for further passivating treatment.
- 4) Deep reactive ion etching (DRIE or RIE) of one or both membrane sides to form construction elements or cavity.
- 5) Anisotropic wet etching of one membrane side to form a cavity.
- 6) Metal sputtering (Al).
- 7) Anodic bonding [64] with borosilicate glass (BF33).
- 8) Sensor packaging.

The described manufacturing route is fully compatible with the SOI technology [21], [39], [40]. The advantages of applying SOI plates on the bases are the increased operational temperature range and increased device resistance. The drawback of implementing the SOI technology can be its higher costs compared to the standard CMOS-wafer.

Depending on the type of absolute or relative pressure sensor, additional operations can be added to vacuum the cavity under the membrane by applying getter [40] to the glass base. Packaging of the pressure sensor is often performed on the borosilicate glass, e.g., BF33 [21] or PYRE 7740 [20], [65]. Further mounting can be implemented on a printed circuit board (PCB) [4], in a TO-8 like packaging [20] or on a ceramic-metal packaging, such as Kovar [46], by adhesive [33]. Such material choice is caused by the necessity of matching the temperature coefficients of expansion between silicon, glass, and the metal-ceramic base, as well as by the features of the bonding operation. The electrical connection is carried out through the contact pads by means of automatic wire bonding. For use in aggressive environments, the pressure sensor manufactured by CMOS technology can be placed

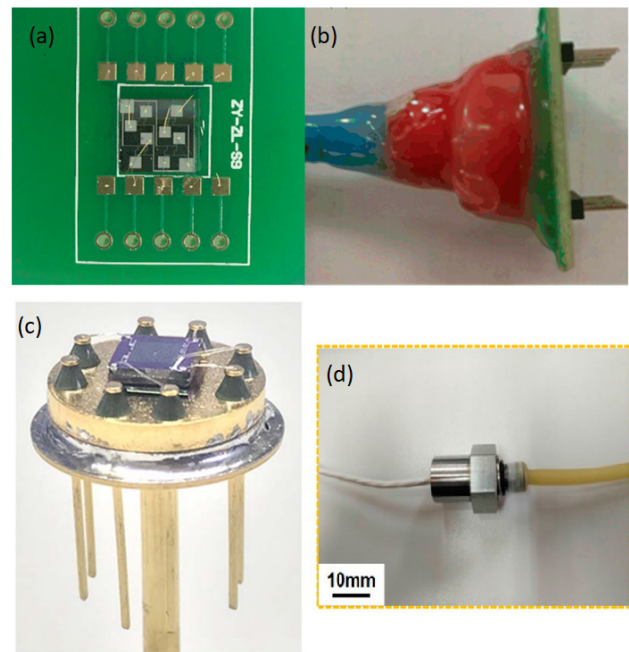


Fig. 7. MEMS pressure sensor final assembly options. (a) Photograph of the pressure sensor on the PCB [4]. (b) Photograph of the packaged pressure sensor [4]. (c) Photograph of the pressure sensor in TO-8 like package without top case part [48]. (d) Photograph of sensor assembled in stainless steel case [21].

in a packaging made of stainless and acid-resistant steel. In this case, an intermediate diaphragm is used and the cavity between the measuring and intermediate diaphragm is filled with specialized oil [65]. Fig. 7 shows the possible options for the final assembly of the pressure sensor inside the packaging.

The article [2] contains the description of the processing route to fabricate MEMS structure by means of applying Minimal-fab technology. The concept of Minimal-fab presupposes using smaller size plates (12.5 mm) for separate technological machines simulating a clean room of a smaller scale. So, the technological procedures that are carried out in such process installations simulate technological procedures which are applied in CMOS-process when fabricating ASIC. The researchers proposed the sensor design based on the CMOS-oscillator. The manufacturing route is presented in Fig. 8.

The main advantages for choosing the concept of Minimal-fab can be lower costs when building prototypes, a higher integration level of micromechanical and microelectronic technological processes, and smaller dependence from the manufacturer. However, currently there are such MEMS ASIC integrations [41], [59], [60], [61] within the typical Mega Fab process, as well as widely available launches within multiproject wafer shuttle from manufacturers, including ASIC and MEMS based on one crystal.

V. MEMS PRESSURE SENSOR PERSPECTIVES

The research works [8], [9], [10], [11], [48], [57], [58], [66] should be described separately. The studies mentioned in [48], [57], and [58] propose integrating bipolar transistors into the SE of a pressure sensor, designed as a membrane with

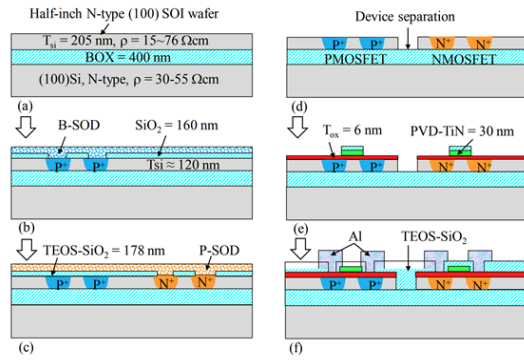


Fig. 8. Manufacturing route of pressure sensor fabrication based on CMOS-oscillator using minimal-fab equipment [2]. (a) Wafer cleaning. (b) Boron diffusion for PMOSFETs using thermal diffusion. (c) Phosphorus diffusion for NMOSFETs using thermal diffusion. (d) Etching of the top Si layer by RIE for the device separation. (e) Gate oxide formation. (f) Contact hole formation by photolithography, wet etching and metallization.

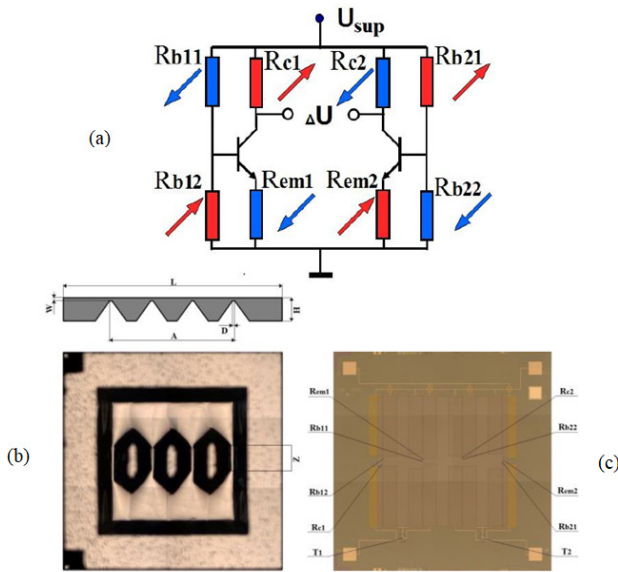


Fig. 9. Pressure sensor chip with a piezosensitive differential amplifier with a negative feedback loop. (a) Schematic view. (b) Back view and schematic for a cut of a chip mechanical part with geometrical indicators. (c) Top view with location indicators of all circuit elements [48].

the piezoresistive bridge to increase sensitivity during small pressure measurements. Fig. 9 shows the electric circuit and microphotography of such a MMPS [48]. The application of transistors and the preamplification stage enabled to double the sensitivity to pressure (with ~ 5 to ~ 11 mV/V/kPa) compared to standard MMPS based on the piezoresistive data registering [19], [25] when maintaining the measurements range of 5 kPa.

MMPS with integrated bipolar transistors [48] has a number of disadvantages: a complicated manufacturing route, including three cycles of photo lithography, doping with different concentrations; the necessity in high-precise matching of piezoresistors topology and the membrane thin areas etched by means of the liquid method.

Gardner et al. [67] offer to apply MOSFET instead of piezoresistors. Carriers' mobility in MOSFET channel changes under mechanical stresses, and consequently, changes the output voltage proportionally to the applied pressure. n-MOSFETs are connected based on the principle of the

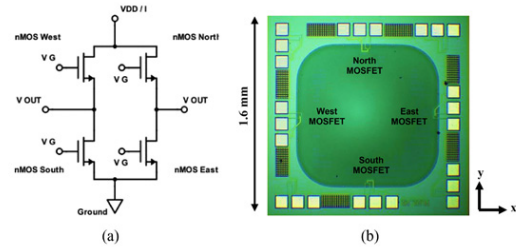


Fig. 10. (a) Schematic showing the operation of the MOSFET pressure sensor for the Wheatstone bridge operation. (b) Optical micrograph of the fabricated sensor. The crystallographic orientation of p-type MOSFET is [110] and follows the x-axis, whereas the n-type MOSFET is [100] and follows the y-axis [67].

Wheatstone bridge (see Fig. 10), and the flowing current can be calculated as follows [67]:

$$I_{DS} = \mu C_{ox} \frac{W}{L} \left(\frac{(V_G - V_T)^2}{2} \right) \quad (14)$$

where I_{DS} is the drain current, μ is the carrier mobility, W and L are the channel length and width, respectively, V_G is the gate voltage, and V_T is the threshold voltage.

Under pressure, the mobility of charge carriers in the transistor channel can change in accordance with the following equation that is proportional to the mechanical stresses in the membrane [67]:

$$\Delta I_{DS}/I_{DS} = \Delta\mu/\mu = \pi_l\sigma_l + \pi_t\sigma_t \quad (15)$$

where $(\Delta I_{DS})/I_{DS}$ is the change in drain current and $\Delta\mu/\mu$ is the change in carrier mobility.

The developers of this solution obtained a significant increase in sensor sensitivity owing to using n-MOSFET instead of p-MOSFET due to high mobility of charge carriers.

A similar design of the bridge to increase the sensor sensitivity based on MOSFET is offered in [9] and [10]; the researchers also proposed the complete methodology to simulate n-MOSFET by applying the FEM analysis in COMSOL, and sending the results to SPICE-simulator, LT-SPICE, and Simulink.

The main disadvantage of using transistors as a part of membrane is smaller linearity of MMPS compared to MMPS based on the Wheatstone bridge thus resulting in n-MOSFET MMPS becoming a compromise when applied in small ranges of pressure measurements.

Articles [11] and [66] deal with the membrane designs consisting of solely piezoresistive materials to simplify the overall design.

Thus, research [11] describes MMPS based on poly-SiC (polycrystal silicon carbide) obtained by means of surface micromachining method (see Fig. 11).

The technological process of sensor membrane fabrication consists of three stages: the first stage deals with forming a sacrificial layer of SiO_2 in the area of further space; the second stage deals with forming the base of a further membrane above the sacrificial layer with release holes; the third stage deals with etching of the sacrificial layer of SiO_2 and sealing release holes on the membrane surface, and the sensor end-pressure can be also formed at this stage.

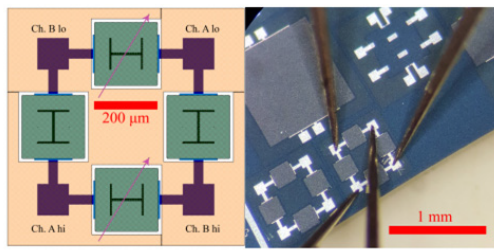


Fig. 11. Wheatstone bridge connections to the source measurement unit; (left) schematically and (right) through microscope [11].

TABLE I
COMPARISON OF PIEZORESISTIVE COEFFICIENTS OF SILICON
AND SILICON CARBIDE [12]

Piezoresistive coefficient 10^{-11} Pa^{-1}	n-Si	p-Si	p- 3C- SiC	n- 3C- SiC	n- Poly- Si	p- Poly -Si
π_{11}	-102.2	6.6	-9.6	1.5	-45.4	58.8
π_{12}	53.4	-1.1	5.8	-1.4	25	27.2
π_{44}	-13.6	138.1	1.6	18.1	-70.4	85.9

To apply as a SE, the membrane was doped with impurities of n-type and combined with four other types to form the Wheatstone bridge. The researchers obtained the logarithmic dependence of the output signal ranging from 10 Pa to 100 kPa with proportional coefficient of 1.1 mB/B/bar.

Patankar et al. [12], Verma et al. [13], Patankar et al. [42], and Tian et al. [43] studied the structures based on SiC as they have a high potential when operating in aggressive and tough environments.

Silicon carbide is a semiconductor with a wide band gap of 3 eV; thus, it possesses bigger temperature stability and better resistance to high-energy radiation, compared to typical Si [11]. The low coefficient of temperature expansion improved mechanical properties, as well as chemical inertia, make SiC an optimal material for applying tough conditions [13].

Owing to the peculiarities of interatomic bonds, there are more than 200 SiC modifications that differ in their characteristics depending on their crystallographic structure. The most widely used polytypes are as follows: 3C-SiC, 4H-SiC and 6H-SiC. SiC polytypes can be either isotropic or anisotropic materials and have different piezoresistivity coefficients [12].

The design methodology and typical designs of MMPS based on SiC are compatible with MMPS methods based on Si; however, the piezoresistivity coefficients of SiC have much lower values (see Table I) resulting in sensor sensitivity reduction. In addition, the Young's modulus and SiC density are 2 times and 1.5 times more, respectively, compared to Si; thus, enabling to design a sensor for bigger maximal mechanical stresses and compensates for the smallest piezoresistive coefficient.

Nowadays, the main disadvantages of MMPS based on SiC are insufficient availability of SiC processing technology, as well as slow development of CMOS-SiC technology, thus making SiC MMPS the most feasible in the future.

Several scientists [14], [15], [44], [45], and [66] described the option to apply polymer membranes [polydimethylsiloxane

(PDMS)/polymethylmethacrylate (PMMA)] and several other materials [aluminum doped ZnO (AZO) and carbon nanotube (CNT)] that are not based on silicon.

In particular, the application of polymer MMPS with the membrane based on PDMS and PMMA [15], [45], as well as MMPS-based AZO films can be developed with a flexible and nontoxic sensor, for example, for biomedical needs. The researchers also focused on the sensors based on capacitive [15] and piezoresistive data registering [45], [66].

One of the most promising technical solutions could be the application of CNTs and graphene MMPS [14], [44] as transistor materials due to high temperature stability in big piezoresistive coefficients compared to Si. The main drawback of such a solution can be a necessity to complicate the technological process compared to standard CMOS-technology.

VI. ICs OF MEMS PRESSURE SENSOR

ICs for a micromechanical sensor consists of an amplifier, analog-to-digital converter (ADC), and digital circuit of signal processing (see Fig. 1).

Generally, modern MMPS contains an ASIC enabling to control and convert the sensor signal. ASICs of a pressure sensor can be designed by means of combining an analog interface (AI) and digital processing system that assists in processing, storing, and transmitting data and be compatible with widely used communication protocols for different applications. The important aspect is correlation of MMPS and ASIC, as when designing a sensor; the main focus is selection of possible combination solutions, aimed at reducing any side effects between two system components placed together inside one case. The minimal distances, materials with minimal sensitivity to external conditions, and optimal components sizes have become the main factors that are taken into account when fabricating a pressure sensor. The type of the applied combination of MMPS and ASIC is determined by its application.

The first interface unit is an AI. The main structural AI unit is an amplifier with one or several amplifying cascades. The amplifier can be fabricated in inverting and noninverting configurations, and has a high common-mode rejection ratio to suppress any possible common-mode interferences that can be transmitted by means of a sensor at AI input from the earth node and feeding voltage power rail. AI must detect an insignificant signal, but it must not load a sensor output; thus, it is necessary to provide high input impedance at AI input. That is why CMOS technology is more preferred to design an amplifier in the input AI cascade. The amplifier is supposed to satisfy the following requirements: a low noise level, high input resistance, and low voltage zero offset.

These requirements can be met by application of the instrumentation amplifier. The most widely used instrumentation amplifier design consists of three operational amplifiers (OpA) and resistive feedback (FB), setting the amplification coefficients [68], [69], [70]. The amplification coefficient of such an instrumentation amplifier can be adjusted by means of resistors' matrix. The design of the instrumentation amplifier based on three OpA's is presented in Fig. 12. The instrumentation amplifier described in [68] has the following characteristics:

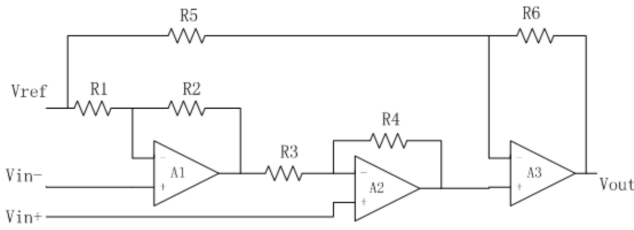


Fig. 12. Schematic of the instrumentation amplifier based on three OAs with resistive FB [68].

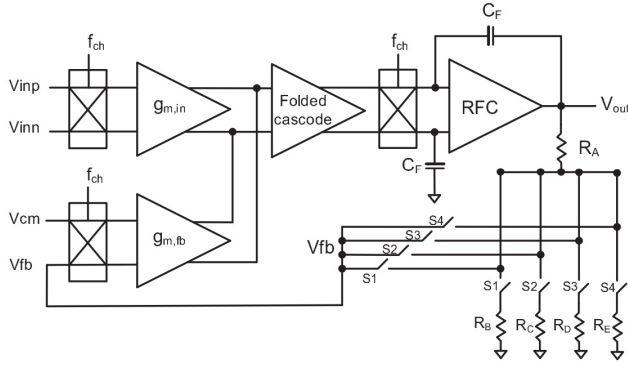


Fig. 13. Schematic of current feedback instrumentation amplifier [72].

the common-mode rejection ratio of 80 dB, the bandwidth of 500 kHz, the adjusted gain of 15–60, and the noise amplitude spectral density supplied to the instrumentation amplifier input is less than $100 \text{ nV}/\sqrt{\text{Hz}}$ with the frequency of 1 Hz.

The research in [71] has provided the design of the current feedback instrumentation amplifier based on AD844 discrete microchips. The integrated design of such instrumentation amplifiers is described, for example, in articles [72] and [73]. The instrumentation amplifier developed in [73] has a low noise level: the noise amplitude spectral density supplied to the instrumentation amplifier input makes up $50 \text{ nV}/\sqrt{\text{Hz}}$ with the frequency of 1 Hz. Also, the advantages of such an amplifier are the low shift voltage of $11.7 \mu\text{V}$ and the low consumed current of $250 \mu\text{A}$. The functional chart of current feedback instrumentation amplifier is presented in Fig. 13.

The studies in [74] and [75] deal with circuits of instrumentation amplifiers with capacitive feedback. The feature of such circuitries is low consumed power of some microwatts. The schematic of the instrumentation amplifier with capacitive feedback is presented in Fig. 14. The instrumentation amplifier described in [75] has low consumed current of $4.5 \mu\text{A}$ and low noise level. The spectral density of noise amplitude supplied to the output makes up $42.66 \text{ nV}/\sqrt{\text{Hz}}$ with the frequency of 100 Hz.

In addition, AI can be realized in switched capacitors. However, instrumentation amplifiers based on switched capacitor-integrated circuits have not been widely applied in interfaces of pressure sensors, as they have insignificant input resistance. Such designs are rather complicated for analysis and mostly subjected to the charge leakage effect. Nevertheless, they can be found in commercial applications of microchips, for example, in MAX1454. Such an approach can be based on discrete and continuous time architecture. The latter can be more preferable, due to improved noise

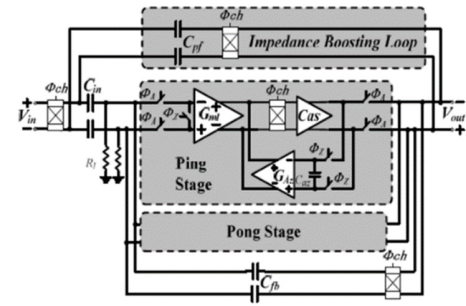


Fig. 14. Functional chart of the instrumentation amplifier with capacitive feedback [75].

characteristics of AI, although it has a lower signal-to-noise ratio. The switched capacitor-integrated circuits are mainly based on the double sampling correlation method to eliminate low-frequency noise suppression.

To decrease flicker noise, the signal modulation–demodulation method (chopping) is applied [72]. To design an input couple of the amplifier cascade, the p-type transistors are applied, as they have a lower level of flicker noises than n-type transistors.

The methods used to match transistors, whether they are interfinger structure type or shared center array to decrease zero shift voltage, are applied when designing amplifiers topology. The research [70] deals with voltage corrective adjustment that is carried out by means of the configured resistor. There are also digital-to-analog converter (DAC) adjustment methods. Moreover, the automated corrective adjustment method is currently used [72], [73], [74], [75].

After defining the applied technology, the design complexity focuses on finding the balance between several key characteristics.

- 1) Low interior noise.
- 2) Low voltage consumption.
- 3) Low input offset voltage.
- 4) Small area on a crystal.

To convert an increased sensor signal into a digital code, an ADC is used. It is necessary to carry out key parameters analysis to determine ADC structure such as conversion speed, linearity, output resolution, and consumed voltage. As pressure sensors need to process low-frequency signals, it is more relevant to apply sigma–delta ($\Delta\Sigma$) modulation ADCs and successive approximation ADCs. $\Delta\Sigma$ -ADCs provide the resolution of up to 24 bits by means of oversampling and shaping effects of quantization noise. By contrast, successive approximation ADCs, as a rule, have smaller resolution than sigma–delta ADC. However, they became more relevant when ADCs with low consumed voltage are required.

Articles [76], [77], [78], [79], [80], [81], [82], and [83] deal with ICs for capacitive pressure sensors. The characteristic feature of capacitive sensors interfaces is low consumed voltage of hundreds of microwatts. The example of a commercial capacitive pressure sensor is ESCP-BMS1 manufactured by ES systems. The sensor case also consists of a microchip being a sensor interface.

Modern IC solutions have the following characteristics: the root-mean-square value of interior noise is less than

TABLE II
COMMERCIALLY AVAILABLE SENSORS

Name	Size, mm	Measured pressure range, MPa	Basic measurement error, %	Additional temperature error, %	IP rating	Power consumption, W	Temperature range, °C
Метран-150TG [87]	98 x 111 x 202	0.003 – 60	±0.065 – 0.2	±0.25	IP66, IP68	0.9	-40 to +85
UPT-67-R [88]	86 x 55 x 138	0.01 – 100	±0.25	±2	IP67, IP68	1	-20 to +85
AIP-10SH [89]	100 x 178	0.25 – 100	±0.1; ±0.2; ±0.5	±0.2	IP65, IP67	0.7;1	-40 to +70
ДДЭ 084 [90]	Ø15,8 x 27,7	4 – 60	±1	-	-	-	-50 to +50
SPT и 13 mm series [91]	Ø 12,62 x 57,7	0.007 – 34.5	±0.25	±0.5	-	-	-40 to +85
NPI-15J-353SH [92]	22,2 x 22,2 x 36	34.5	±0.075; ±0.050	±0.2	-	-	-40 to +125
RPS/DPS 8000 [93]	25 x 84	0.02 – 7	< 0.01	< 0.02	IP65, IP68	-	-40 to +85

320 μV , the measured pressure range makes up 0–100 kPa, and hysteresis characteristics determining the offset voltage is less than 4.1% [84]. Thus, the consumed voltage makes up hundreds of milliwatt [85].

VII. COMMERCIAL USAGE

MMPSs are among the most important MEMS devices and are widely used industrially. According to Yole Development [86], MMPS market amounted to \$1.7B in 2019 with medical, consumer, industrial, automotive, defense, and aerospace applications. The market expected to grow to \$2.2B in the year 2026. The characteristics of various commercially available MMPSs are presented in Table II.

VIII. CONCLUSION

The provided analysis shows that the main approaches to design SEs of MEMS pressure sensors are based on analytical design methods that allow the designer to obtain rapid evaluation results and FEM using *COMSOL Multiphysics*, *ANSYS* software platforms, and packages to develop micromechanical devices, e.g., *Coventor* and *Intellisuite*.

The application of design techniques makes it possible to determine the optimal output characteristics of the sensor, taking into account the manufacturing technology, materials of functional elements, and the type of useful information retrieval.

The design of MEMS pressure sensors as system-on-chip is associated with the development of ASIC and the integration of component parts through packaging solutions within a single package.

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